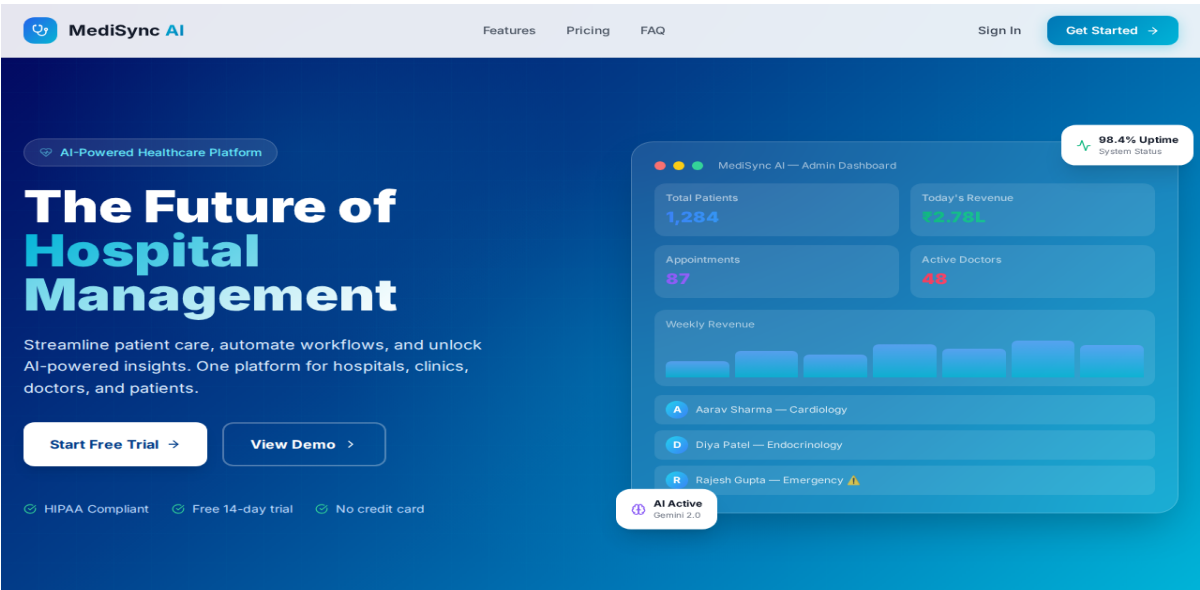


MediSync AI

Hospital Management System User Manual & Visual GUI Guide

Step-by-Step Operational Workflows, Role-Based Portals, and GUI Visuals



Version: 1.0.0 (Enterprise)	System Release: MediSync Web Suite
Audience: All Hospital Roles	Security Level: Confidential & Internal
Updated: May 2026	Platform: Full-Stack React / Node.js

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1. System Overview & Architecture

MediSync AI is a multi-role Hospital Management System designed to optimize clinical workflow, automate front-desk intake, unify e-prescribing, and empower patients through self-service capabilities.

- **Frontend Client:** React 18, TypeScript, Vite, Tailwind CSS, Recharts, Lucide Icons.
- **State Management:** Zustand store with local and API persistence.
- **Backend API:** Node.js, Express, TypeScript, JWT Authorization.
- **Database:** PostgreSQL / SQLite fallback.

2. Getting Started & Account Registration

Creating a New Account:

1. Open `http://localhost:5173/register` in your web browser.
2. Enter your full name, email address, password, and select your role.
3. Click **Create Account** to log in.

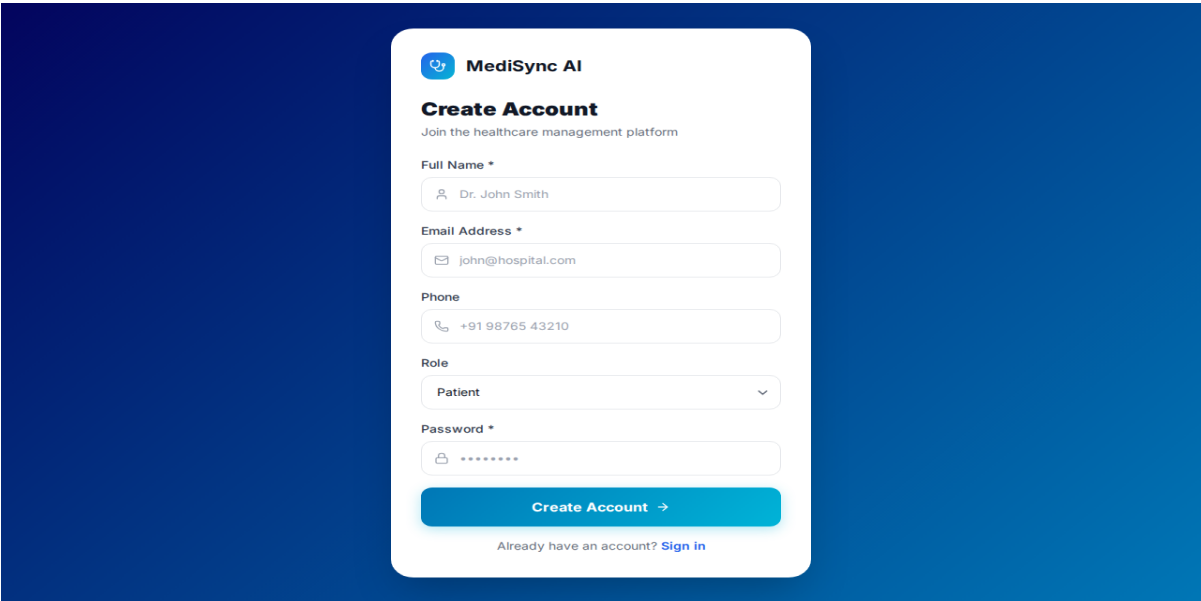


Figure 1: MediSync AI Account Registration GUI Page

Logging In:

Navigate to `/login` and select any Quick Demo Login role badge or enter email/password.

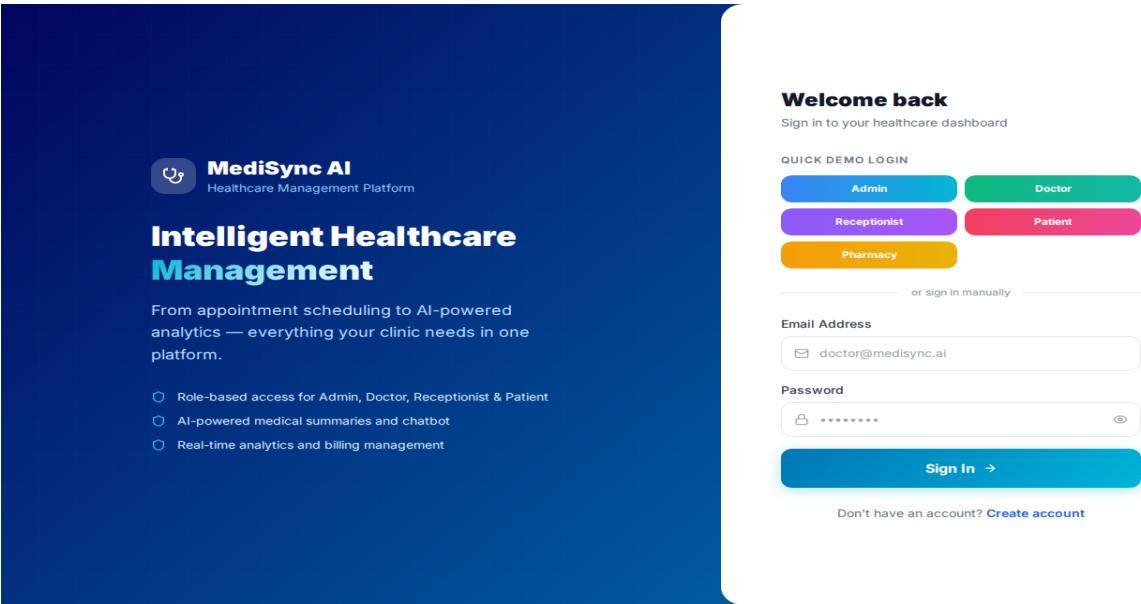


Figure 2: MediSync AI User Login GUI Page

Role	Email Address	Password
Admin	admin@medisync.ai	admin123
Doctor	doctor@medisync.ai	doctor123
Receptionist	receptionist@medisync.ai	recept123
Patient	patient@medisync.ai	patient123
Pharmacy	pharmacy@medisync.ai	pharmacy123

3. Admin Portal Guide

The Admin Dashboard provides real-time oversight of hospital KPIs, patient directories, doctor scheduling, ward beds, billing, and analytics.

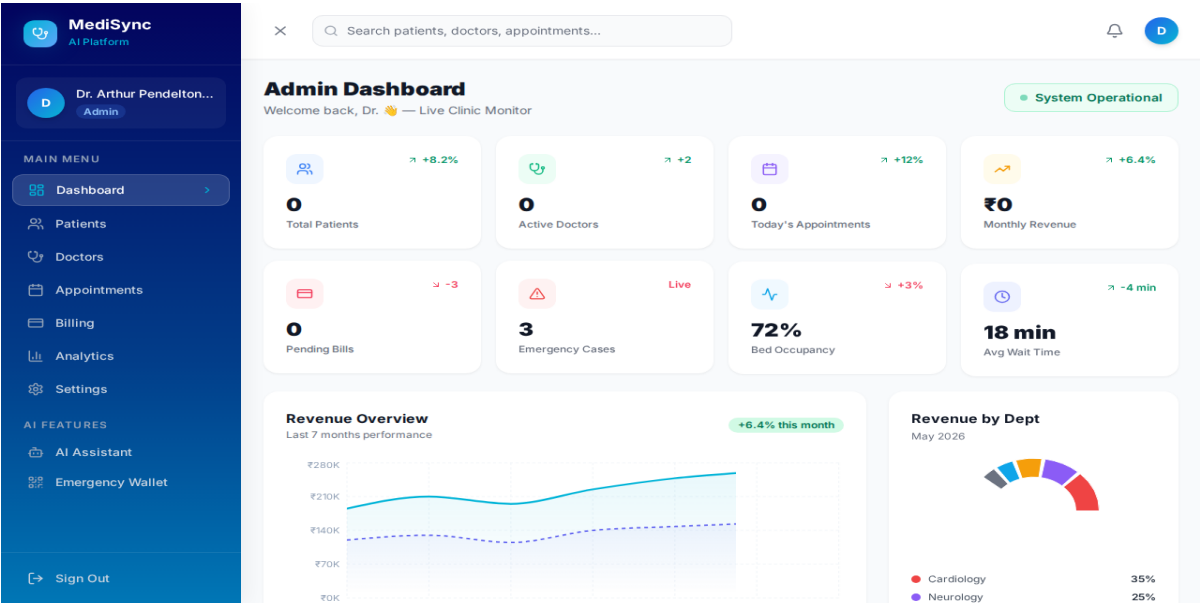


Figure 3: Admin Executive Dashboard Overview GUI

Managing Patient Directory:

Click **Patients** on the sidebar (`/dashboard/admin/patients`) to search, filter by condition, or onboard new patient profiles.

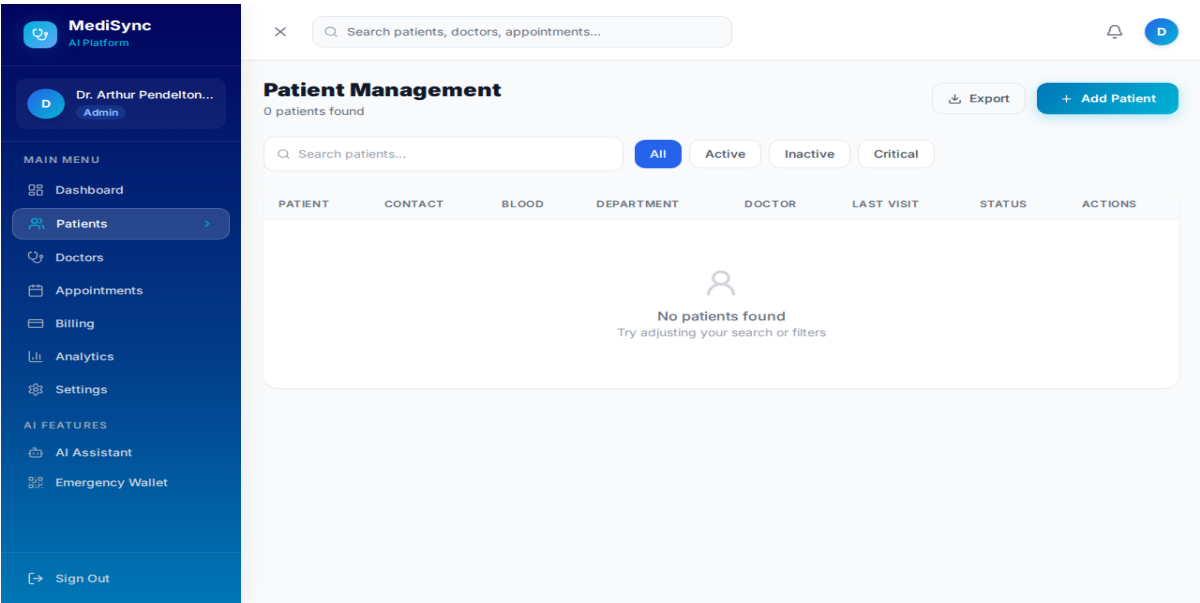


Figure 4: Admin Patient Management & Directory GUI

Doctor & Staff Onboarding:

Click **Doctors** on the sidebar (`/dashboard/admin/doctors`) to onboard new physicians and set consultation fees.

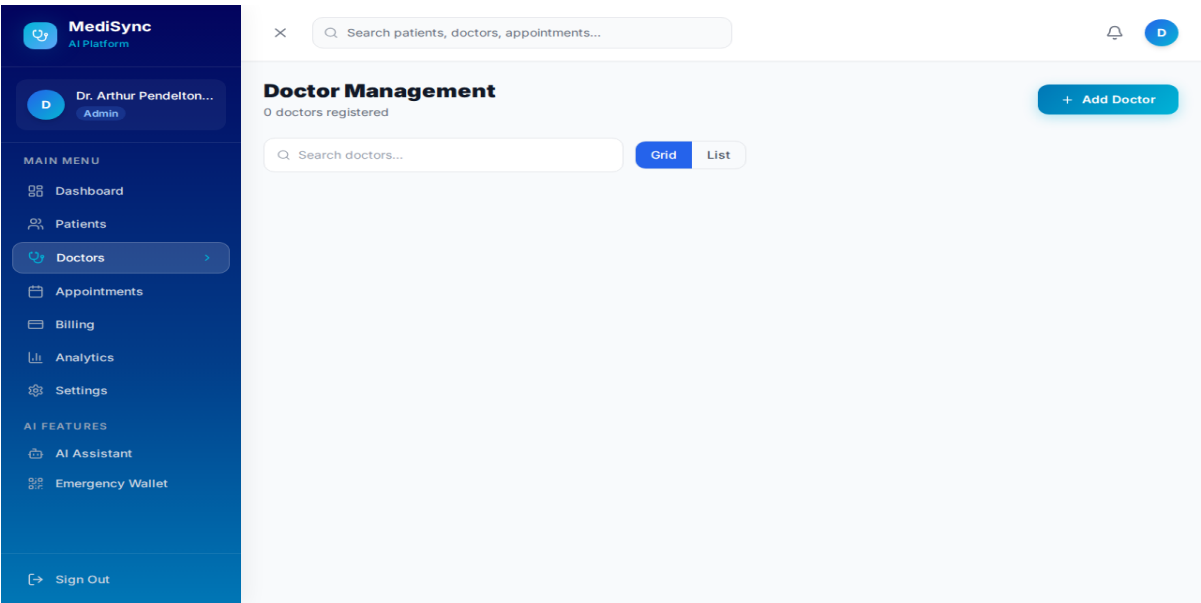


Figure 5: Admin Doctor Directory & Onboarding GUI

Hospital Ward & Bed Management:

Navigate to **Bed Management** (`/dashboard/admin/beds`) to track real-time bed occupancy across ICU, General, and Cardiology wards.

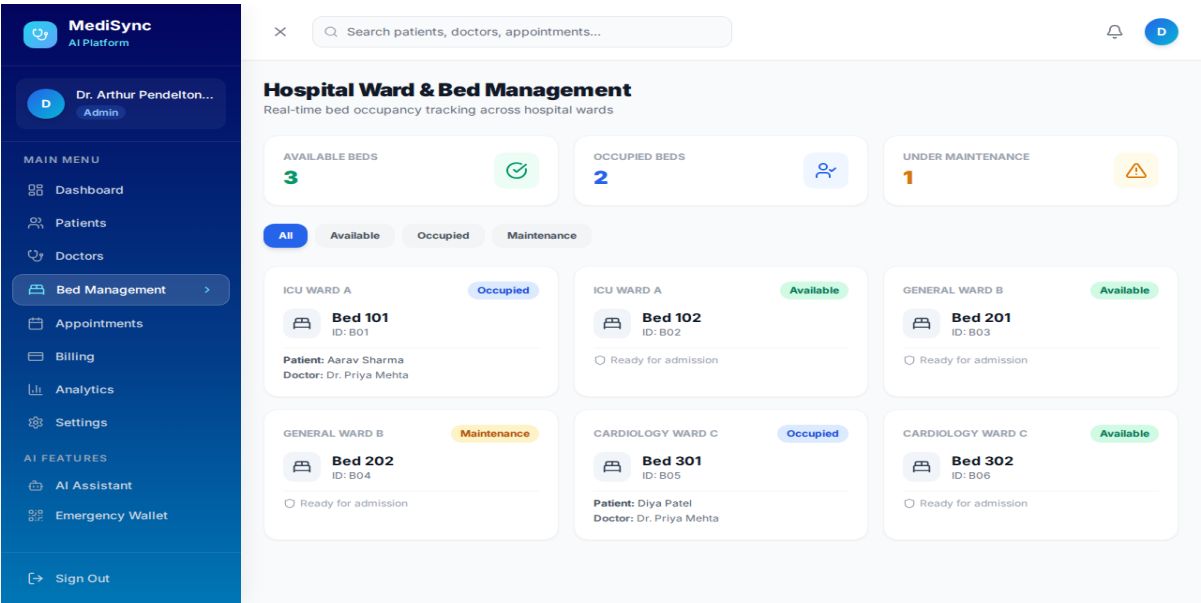


Figure 6: Admin Ward & Bed Management GUI

Billing Oversight & Invoices:

Navigate to **Billing** (`/dashboard/admin/billing`) to inspect itemized bills and payment states (`Paid`, `Pending`, `Overdue`).

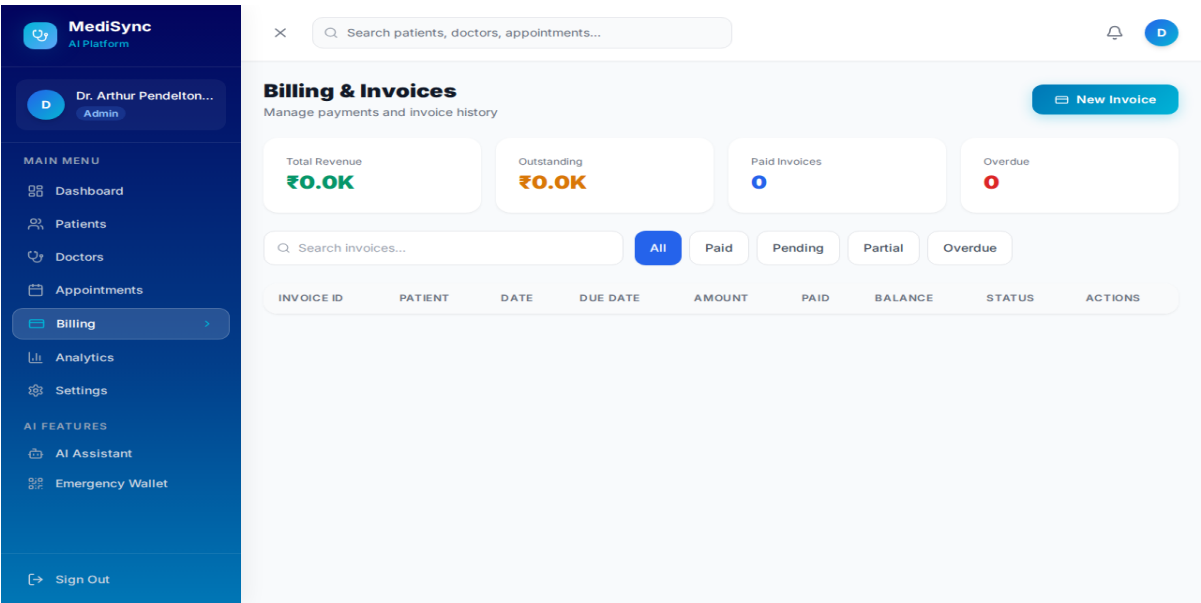


Figure 7: Admin Billing & Financial Invoices GUI

Analytics & Performance Metrics:

Navigate to **Analytics** (`/dashboard/admin/analytics`) to view visual revenue growth trends and bed occupancy rates.

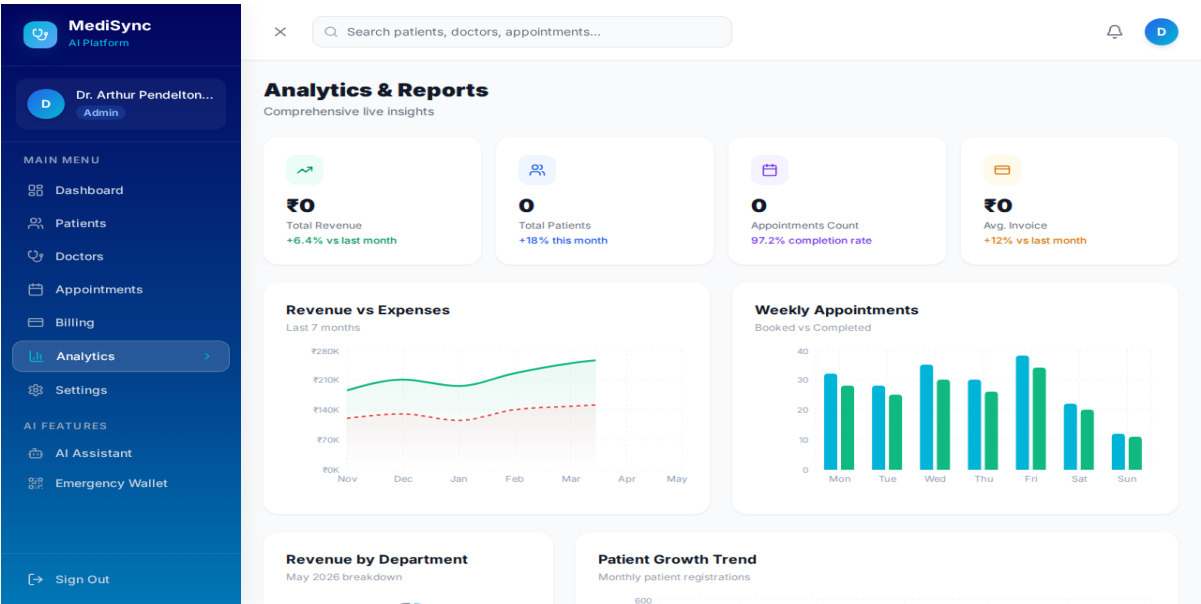


Figure 8: Admin Analytics & Recharts Performance Dashboard GUI

System Settings:

Navigate to **Settings** (`/dashboard/admin/settings`) to configure hospital profile parameters and backup options.

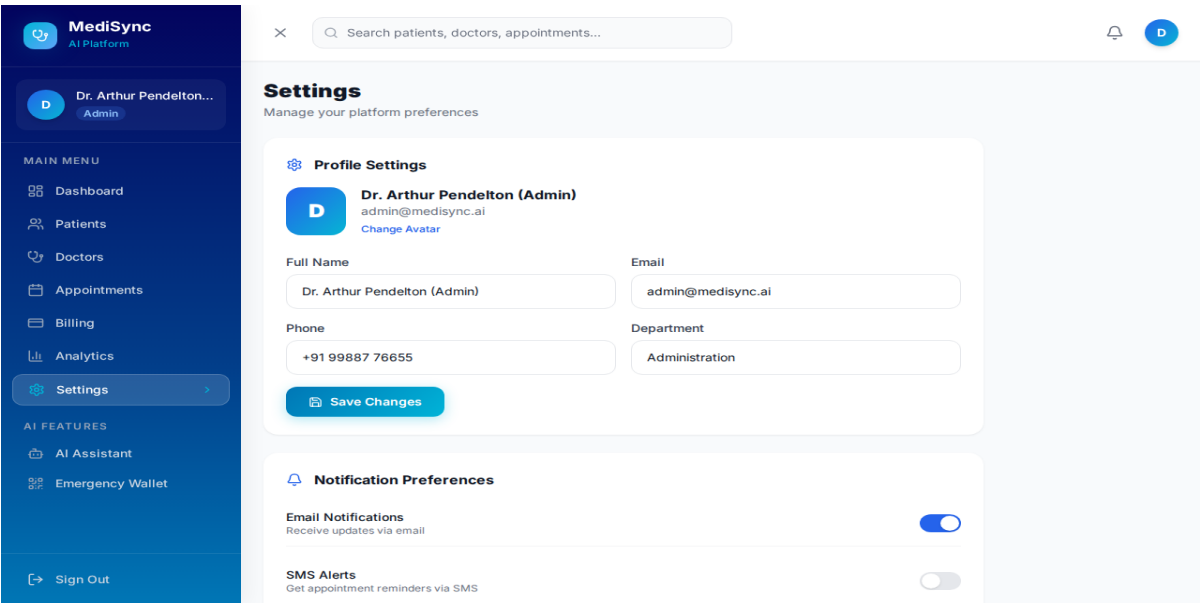


Figure 9: Admin System Settings & Configurations GUI

4. Doctor Portal Guide

Clinicians use this portal to manage patient consultations, review medical histories, launch virtual telehealth sessions, and issue signed e-prescriptions.

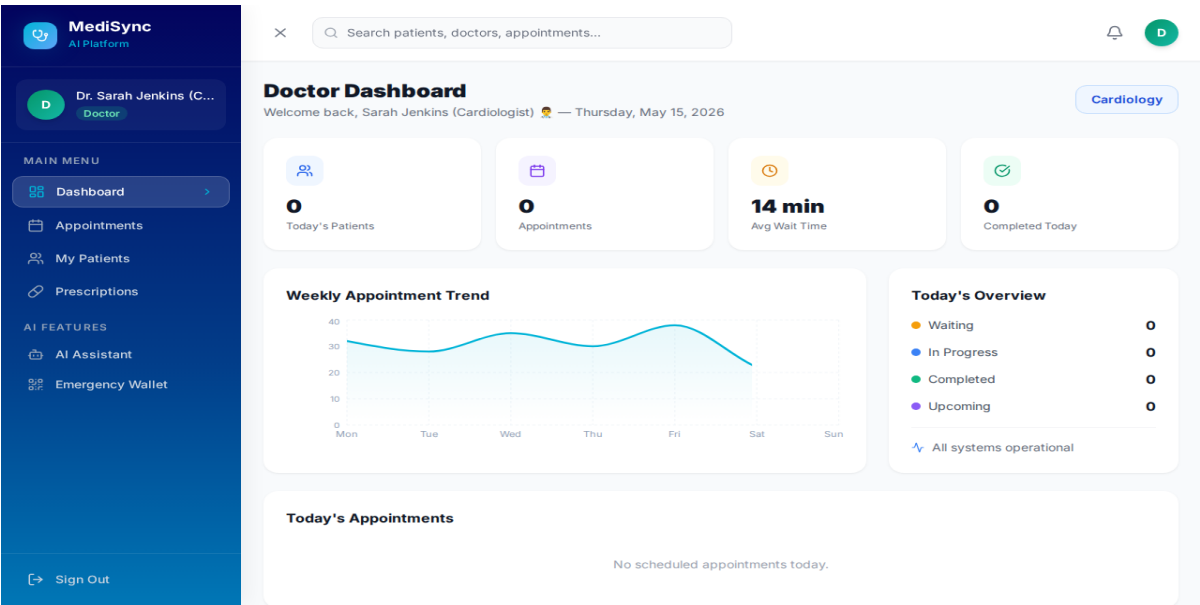


Figure 10: Doctor Consultation Overview Dashboard GUI

Doctor Appointments Desk:

Navigate to **Appointments** (`/dashboard/doctor/appointments`) to update patient lifecycle status (`In Progress`, `Completed`).

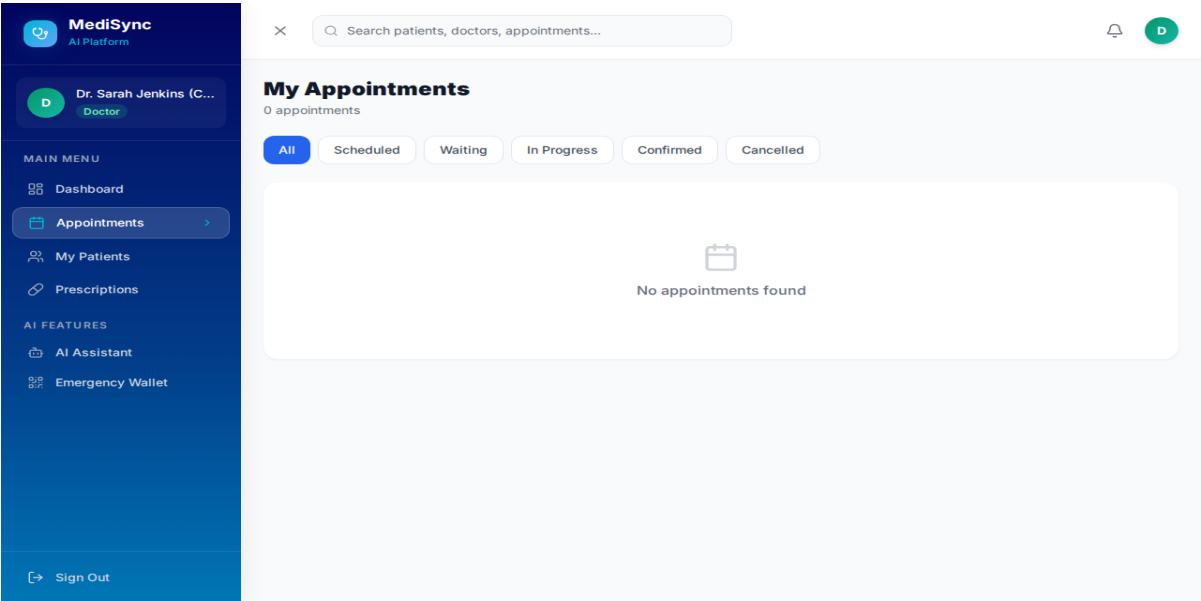


Figure 11: Doctor Consultation Appointments Schedule GUI

Telehealth Virtual Consultations:

Navigate to **Telehealth Consult** (`/dashboard/doctor/telehealth`) to conduct HD virtual video consultations with remote patients.

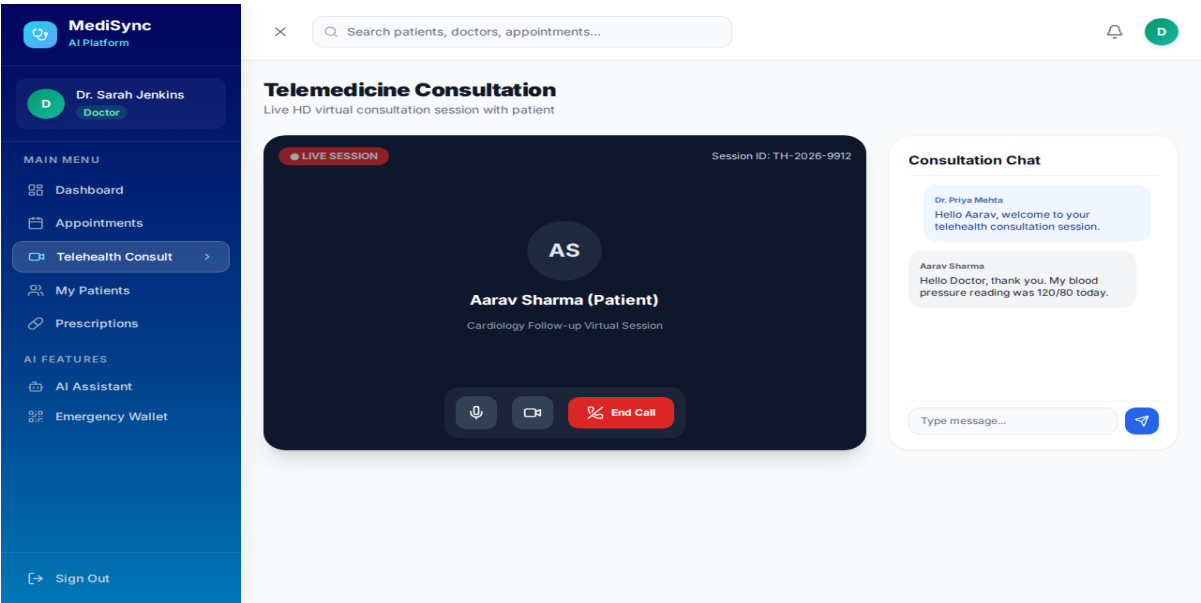


Figure 12: Doctor Telehealth Video Session GUI

Patient EHR Records:

Navigate to **Patients** (`/dashboard/doctor/patients`) to inspect historical vital sign trends and diagnostic notes.

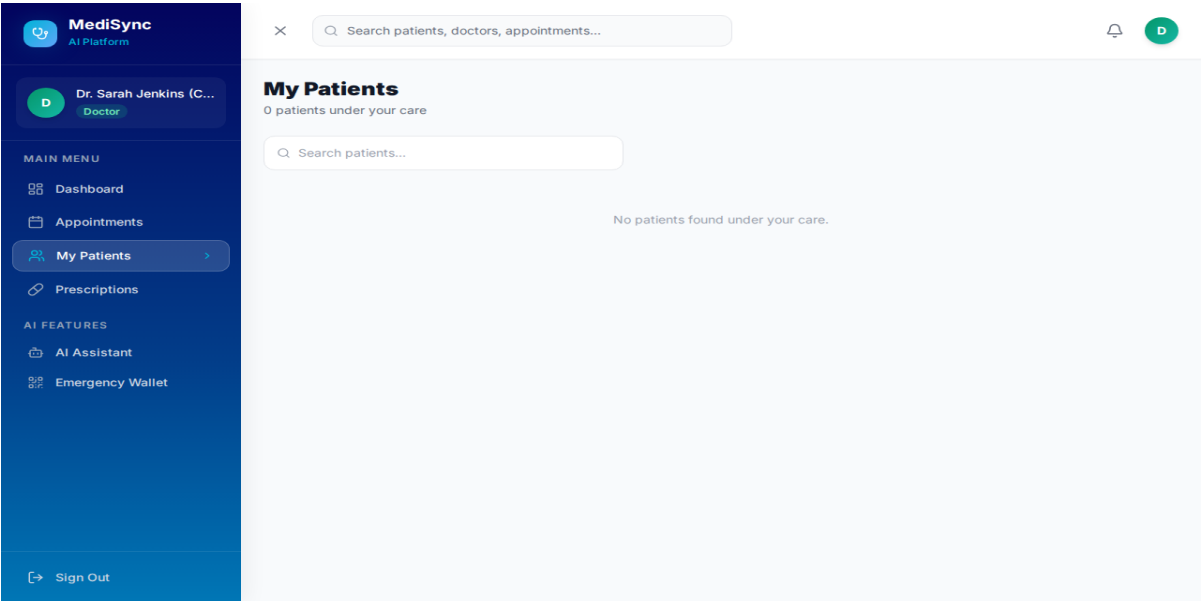


Figure 13: Doctor Patient EHR Health Records GUI

e-Prescribing Workflow:

Navigate to **Prescriptions** (`/dashboard/doctor/prescriptions`). Click **New Prescription**, select patient, specify drug names, dosages, and frequency. Digital signatures are automatically stamped.

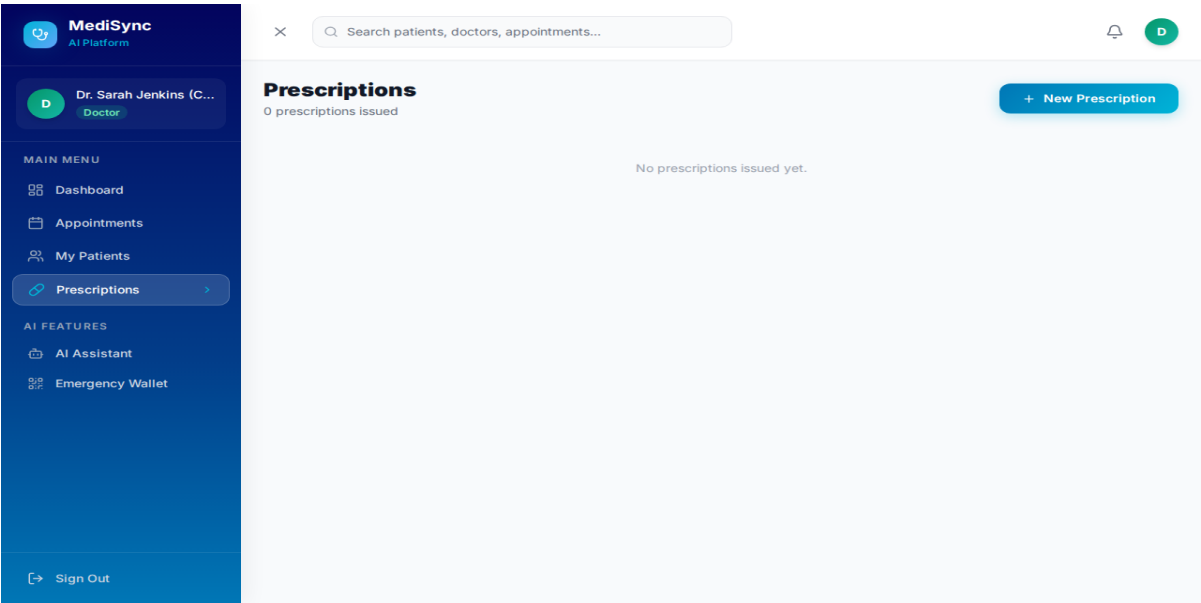


Figure 14: Doctor Electronic Prescribing Interface GUI

5. Receptionist Portal Guide

Designed for front-desk triage, walk-in patient registration, appointment scheduling, and consultation queue optimization.

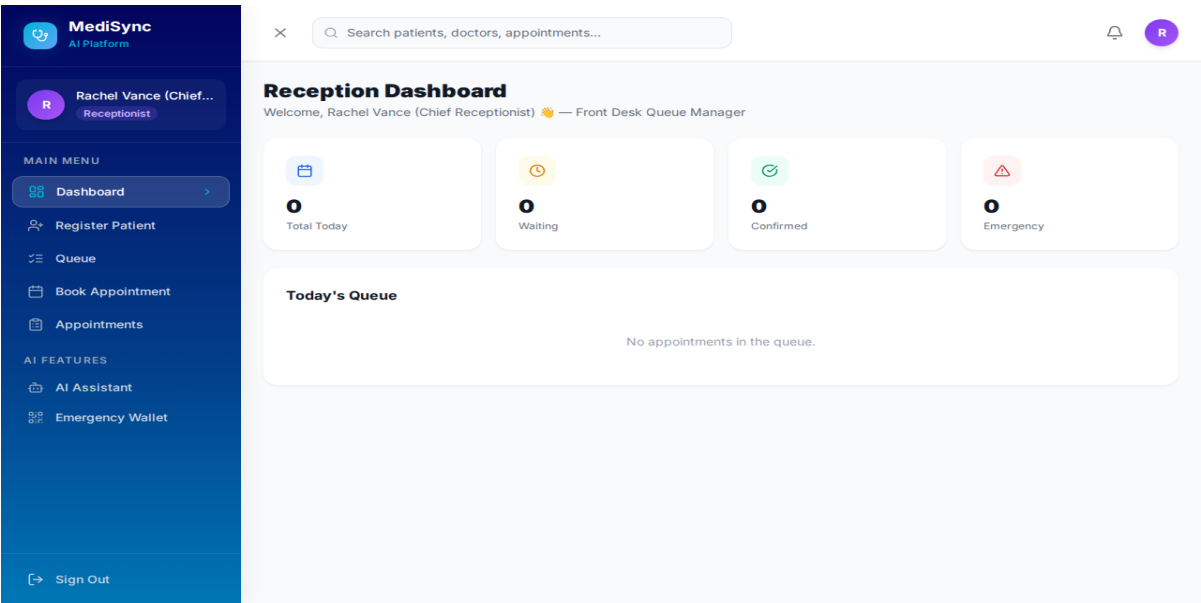


Figure 15: Receptionist Overview Dashboard GUI

Walk-In Patient Registration:

Navigate to **Register Patient** (`/dashboard/receptionist/register-patient`). Fill in vital contact info, blood group, emergency contact, and insurance details.

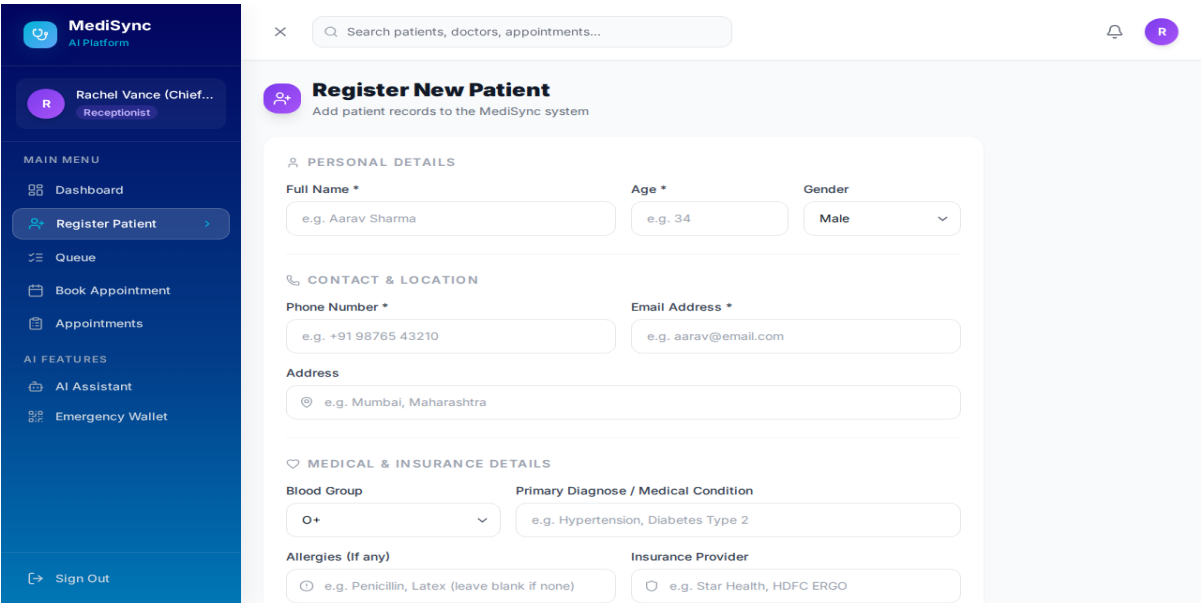


Figure 16: Receptionist Patient Intake Form GUI

Appointment Booking Desk:

Navigate to **Book Appointment** (`/dashboard/receptionist/book`). Select patient, physician, date/time slot, and set priority level.

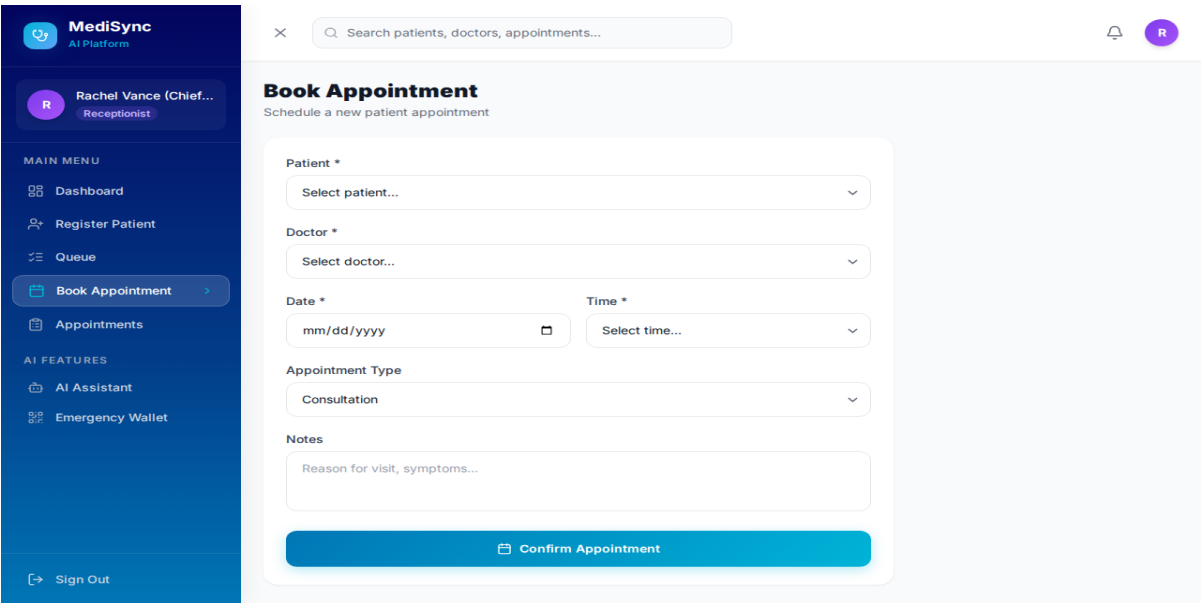


Figure 17: Receptionist Appointment Scheduling Desk GUI

Digital Token Queue Management:

Navigate to **Queue Management** (`/dashboard/receptionist/queue`) to issue sequential consultation tokens and route walk-in patients.

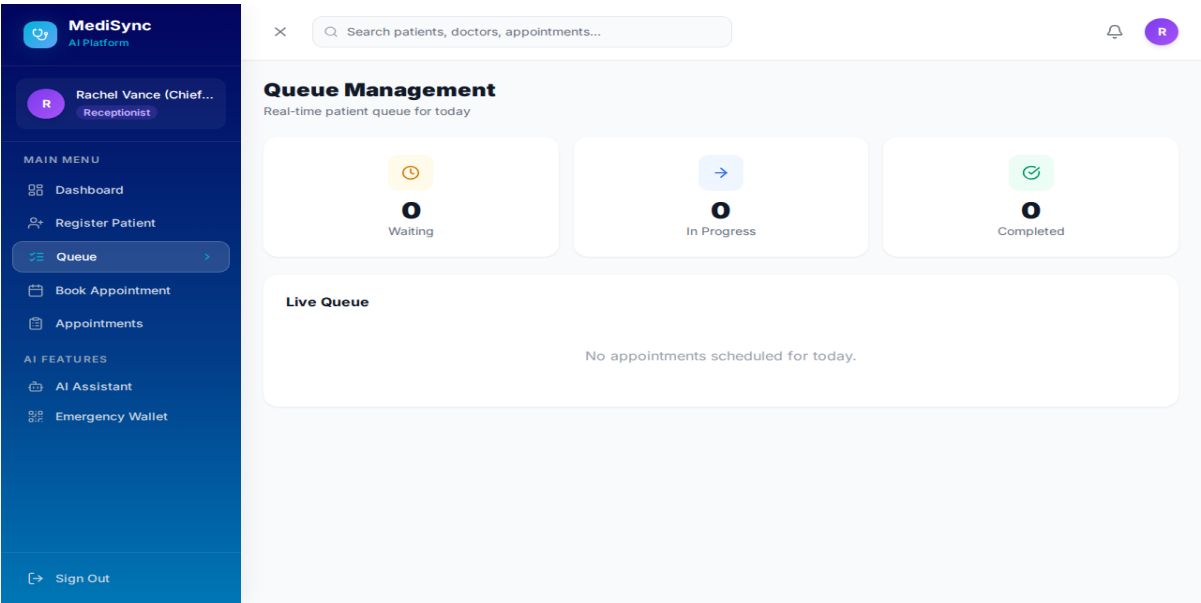


Figure 18: Front Desk Queue & Token Management GUI

6. Patient Portal Guide

Empowers patients to manage their health journey with online booking, medical history access, vitals tracking, and bill payments.

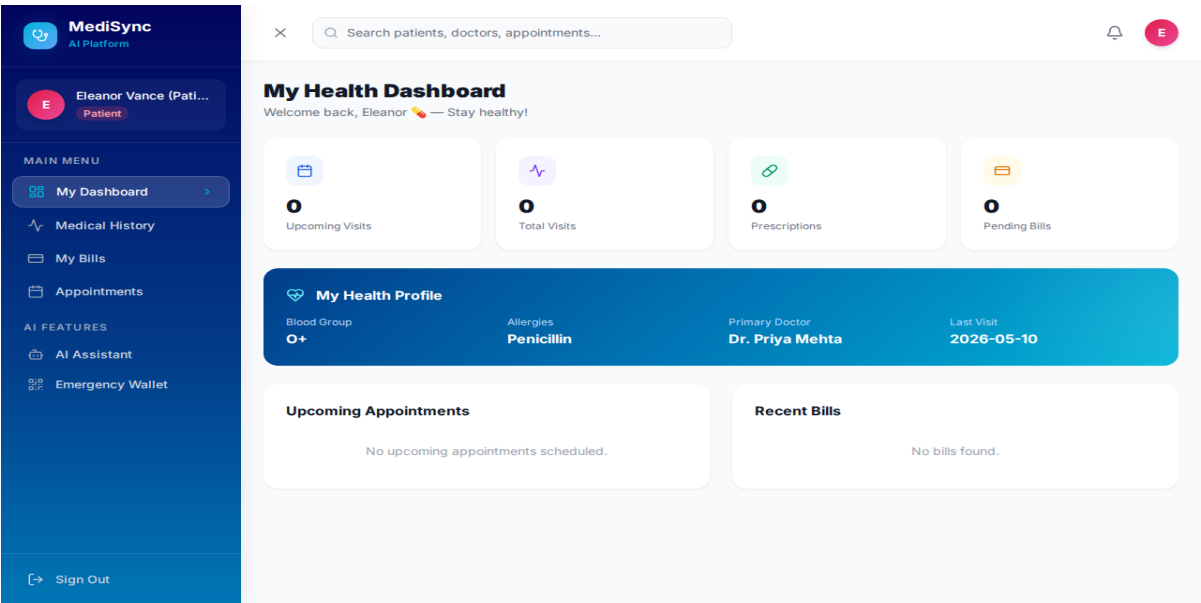


Figure 19: Patient Self-Service Dashboard GUI

Medical History & Prescription Vault:

Navigate to **Medical History** (`/dashboard/patient/history`) to inspect consultation records, past diagnoses, and download signed e-prescriptions.

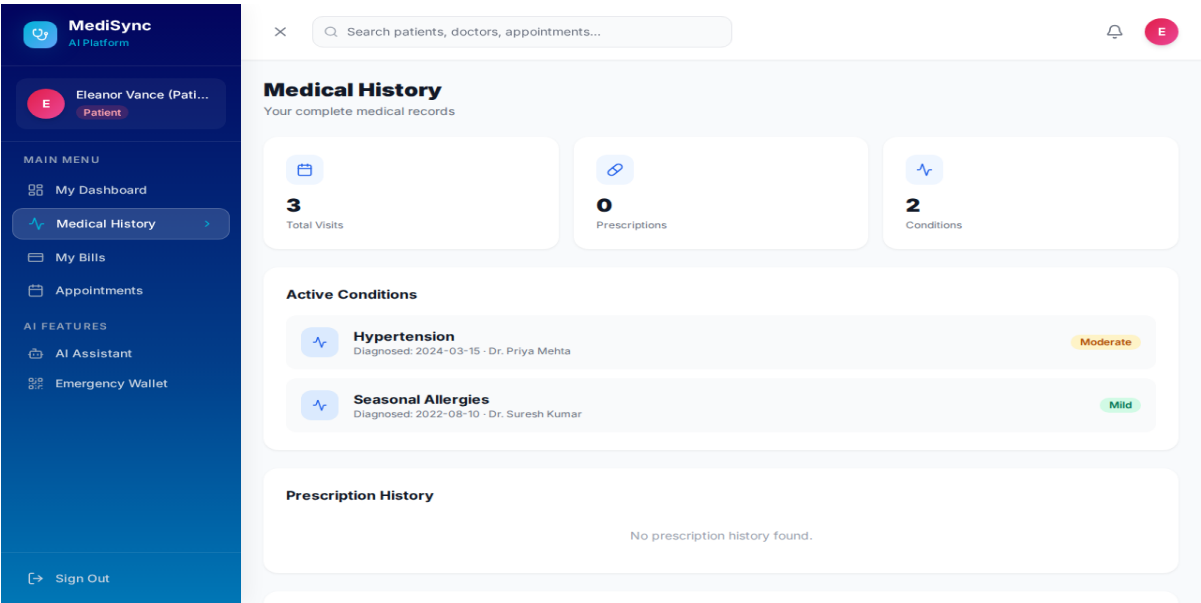


Figure 20: Patient Medical History & Prescription Vault GUI

Patient Vitals & Biometrics Tracker:

Navigate to **Vitals Tracker** (`/dashboard/patient/vitals`) to log blood pressure, heart rate, body temperature, and SpO2 levels.

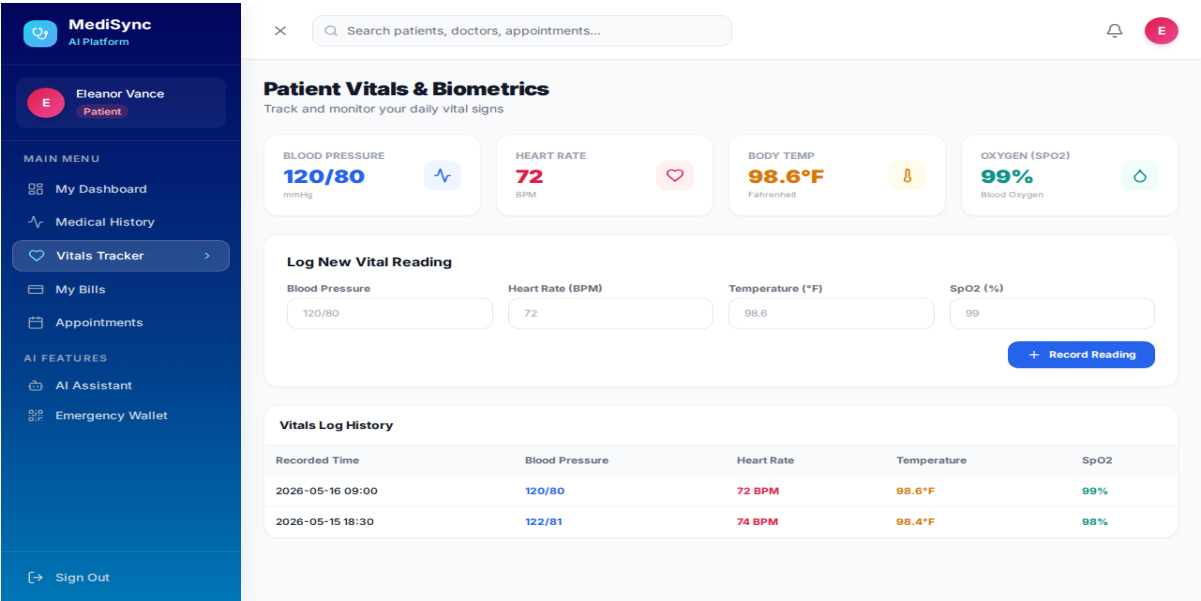


Figure 21: Patient Vitals & Biometrics Tracker GUI

Digital Bill Payments:

Navigate to **My Bills** (`/dashboard/patient/bills`) to view itemized charges and settle invoices online.

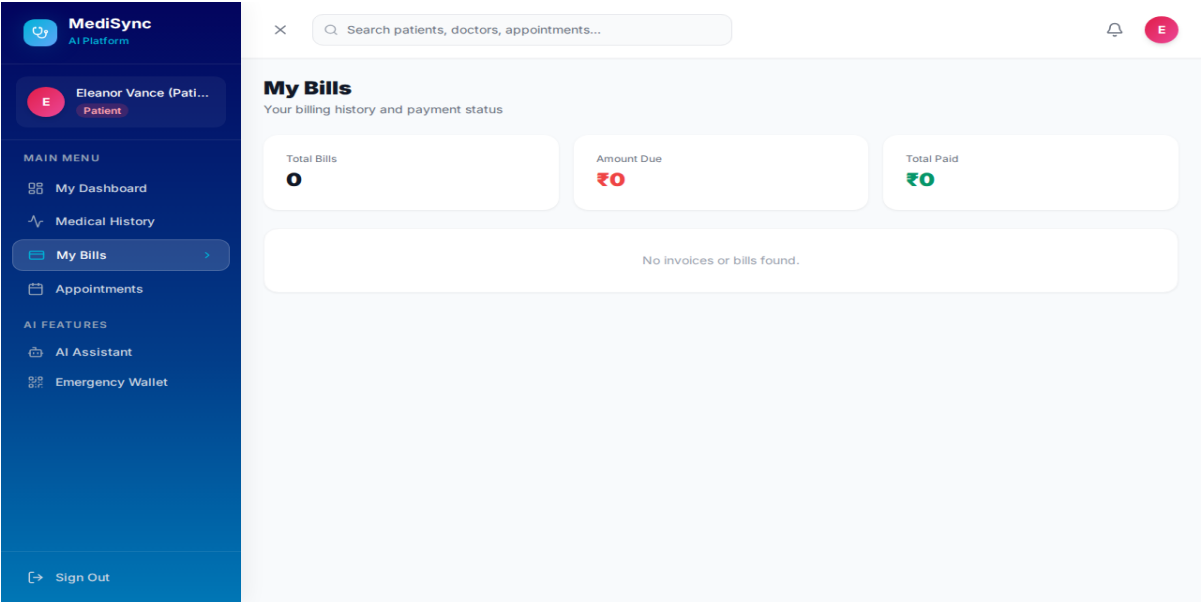


Figure 22: Patient Digital Invoices & Payments GUI

7. Pharmacy Portal Guide

Connects doctors with pharmacists for real-time prescription verification and drug dispensing.

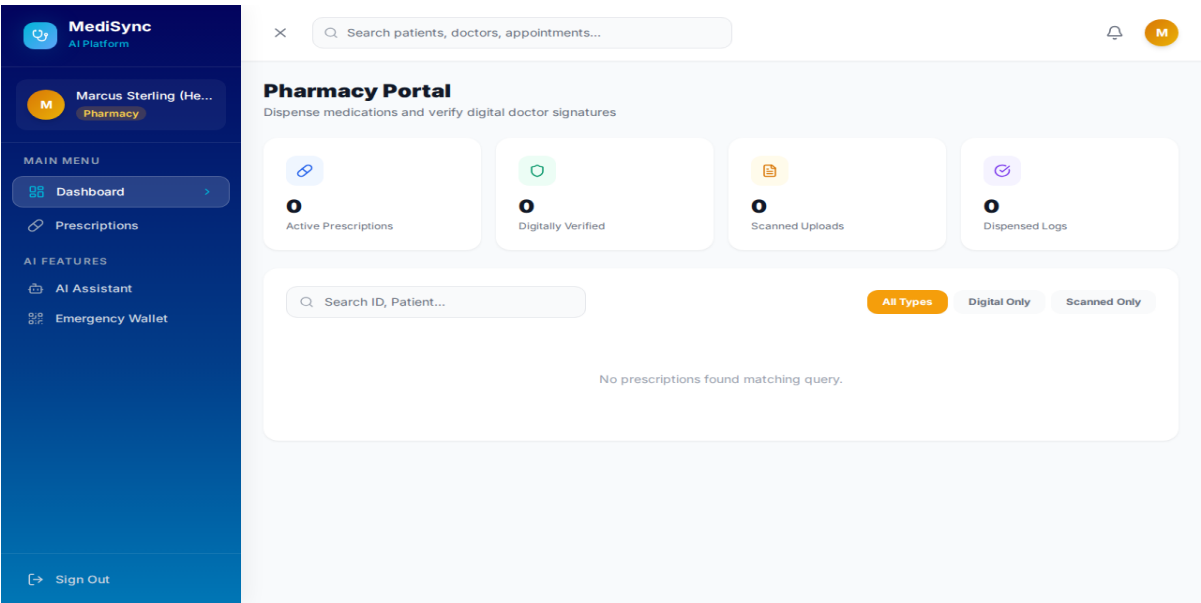


Figure 23: Pharmacy Prescriptions Queue & Dispensing GUI

8. Shared AI & Emergency Tools

MediSync AI Clinical Assistant:

Enter patient symptoms to generate intelligent differential diagnosis insights and formatted summaries.

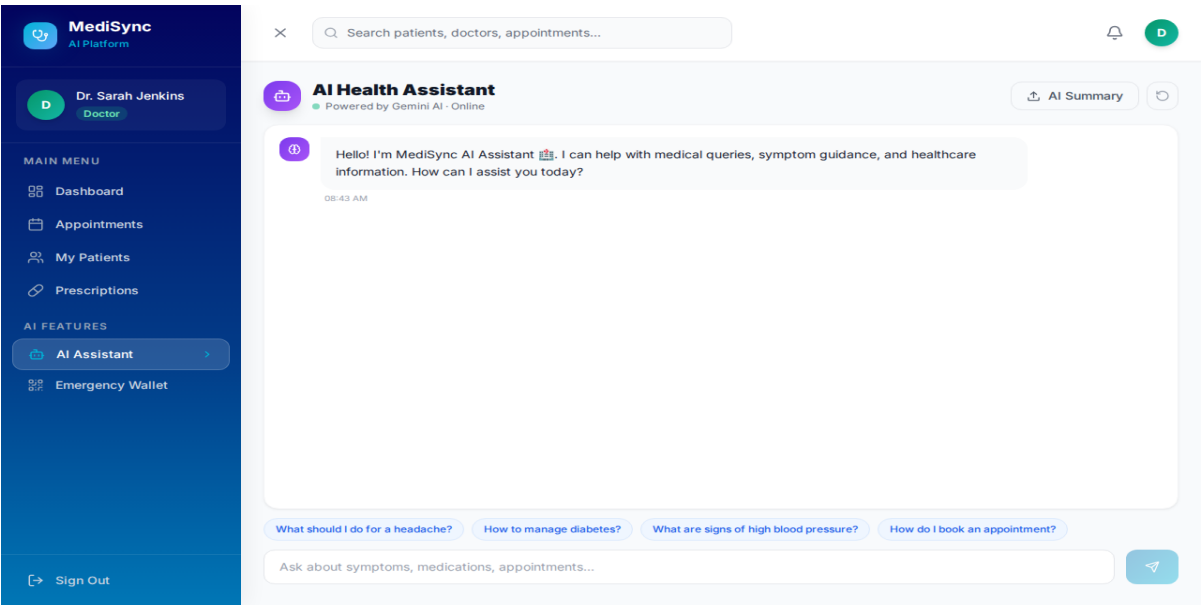


Figure 24: MediSync AI Assistant GUI

Emergency Medical Wallet:

Provides immediate access to critical emergency data: blood group, severe allergies, and emergency contact details.

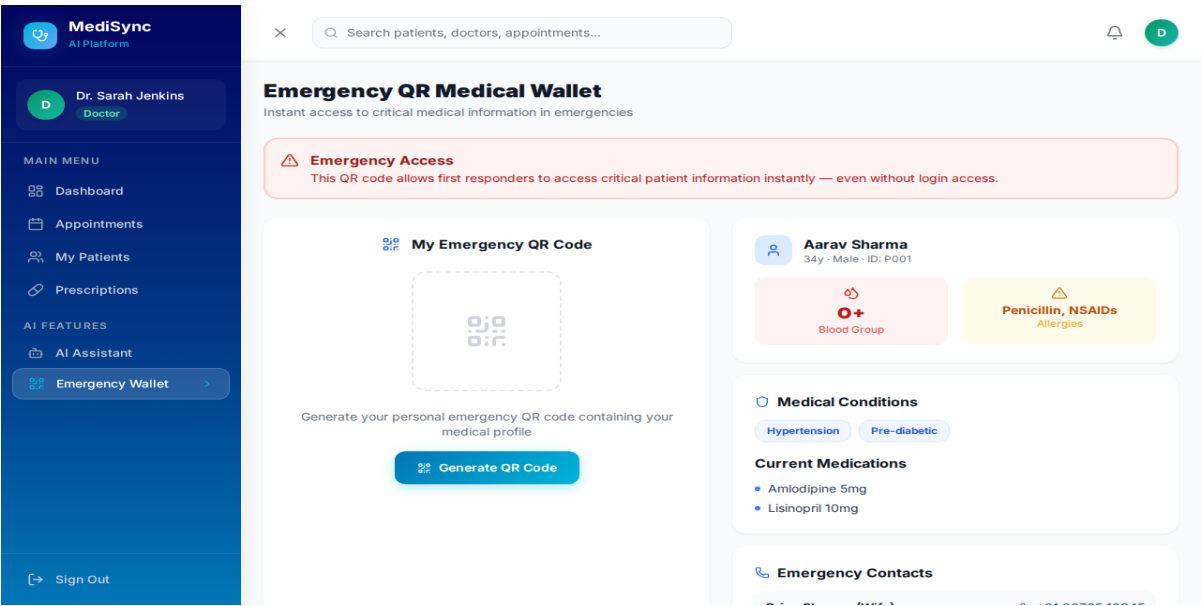


Figure 25: Emergency Medical Wallet GUI

9. API Specifications & Integration

REST API endpoints communicate using JSON format and require JWT Bearer authentication headers.

Endpoint	Method	Description
/api/auth/login	POST	User authentication & JWT generation
/api/patients	GET / POST	Fetch patients list or register new patient
/api/doctors	GET / POST	Fetch doctors directory or add doctor
/api/appointments	GET / POST	Fetch appointments or schedule new session
/api/prescriptions	GET / POST	Fetch prescriptions or generate new e-prescription
/api/invoices/:id/pay	POST	Process digital payment for invoice
/api/beds	GET / PUT	Fetch or update ward bed allocation status
/api/vitals	GET / POST	Fetch or record patient vitals biometrics

10. Troubleshooting & FAQ

Q: How are emergency appointments prioritized?

A: Setting appointment priority to **Emergency** immediately places the patient at the top of the doctor queue and broadcasts system-wide notifications.

Q: Where are digital signatures attached?

A: Doctors upload their digital signature in Profile Settings. The system stamps this signature automatically onto generated e-prescriptions.

